Probability

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1 Set Formulas

$$
\begin{aligned}
P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\
P(A') &= 1 - P(A) \\
(A \cup B)' &= A' \cap B' \\
(A \cap B)' &= A' \cup B'.
\end{aligned}
$$

2 Conditional Probability

$$\boxed{P(E \mid F) = \frac{P(E \cap F)}{P(F)}} \qquad P(F) \neq 0.$$

$$P(F \mid E) = \frac{P(E \cap F)}{P(E)}, \qquad P(E) \neq 0.$$

$$P(E \cap F) = P(E)P(F \mid E) = P(F)P(E \mid F).$$

3 Independent Events

Two events are independent if the occurrence of one does not change the probability of the other.

$$P(E \cap F) = P(E)P(F).$$

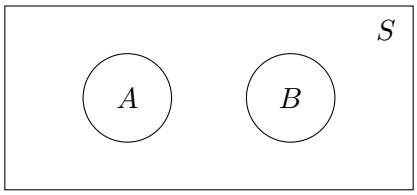
When the conditioning event has positive probability,

$$P(E \mid F) = P(E), \qquad P(F \mid E) = P(F).$$

4 Mutually Exclusive and Exhaustive Events

Mutually exclusive events cannot occur together:

$$A \cap B = \varnothing, \qquad P(A \cap B) = 0.$$



Exhaustive events cover the sample space: $\bigcup_i E_i = S$. If they are also mutually exclusive,

$$\sum_i P(E_i) = 1.$$

5 Theorem of Total Probability

Let $E_1, E_2, \ldots, E_n$ form a partition of S with $P(E_i) > 0$. For any event A,

$$\boxed{P(A) = \sum_{i=1}^{n} P(E_i)P(A \mid E_i).}$$

5.1 Chocolate Store Example

Suppose chocolate is bought from store A or store B. Let E_1 mean store A, E_2 mean store B, and C mean the chocolate is bad. Then

$$P(C) = P(E_1)P(C \mid E_1) + P(E_2)P(C \mid E_2).$$

6 Bayes' Theorem

For a partition $E_1, \ldots, E_n$ and observed event A,

$$\boxed{P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_{j=1}^{n} P(E_j)P(A \mid E_j)}}.$$

For the chocolate example, the probability the chocolate came from store B given it was bad is

$$P(E_2 \mid C) = \frac{P(E_2)P(C \mid E_2)}{P(E_1)P(C \mid E_1) + P(E_2)P(C \mid E_2)}.$$